ALFA CENTER

The 140-hour AI engineer bootcamp

Week 1: Python, math, and ML foundations (Days 1–7, 35 hours)

Days 1–2: Python for AI (10 hours)

Main idea is to be have Python *su ciency* for ML work. Use Kaggle's integrated pathway for zero-setup, ML oriented Python learning.

Day 1 (5 hrs)

Time	Activity	Resource
0:00–2:30	Kaggle Learn Python (Lessons 1–4: syntax, functions, conditionals, lists)	https://www.kaggle.com/learn/python
2:30–4:00	Kaggle Learn Python (Lessons 5–7: loops, strings, dicts + exercises)	https://www.kaggle.com/learn/python
4:00–5:00	Google's Python Class - string and list exercises	https://developers.google.com/edu/python

Day 2 (5 hrs)

Time	Activity	Resource
0:00–2:30	Kaggle Learn Pandas (all 6 lessons - DataFrames, indexing, groupby, dtypes)	https://www.kaggle.com/learn/pandas
2:30–4:00	Kaggle Learn Data Visualization (Matplotlib/Seaborn basics, 4 lessons)	https://www.kaggle.com/learn/data-visualization
4:00–5:00	NumPy crash course - arrays, broadcasting, vectorization (freeCodeCamp YouTube segment + code base)	https://www.freecodecamp.org/news/python-data-science-course-matplotlib-pandas-numpy/ and https://github.com/datapublishings/Course-python-data-science

Supplementary resources (for those who want more):

- Harvard CS50P full course (16 hrs, exceptional teaching):
- https://cs50.harvard.edu/python/ Corey Schafer's Python tutorials: https://www.youtube.com/@coreyms

Days 3–4: Math intuition for ML (10 hours)

just pure visual intuition. 3Blue1Brown is the single best math resource for beginners

Day 3 (5 hrs) - Linear algebra + calculus

Time	Activity	Resource
0:00–1:45	3Blue1Brown "Essence of Linear Algebra" (all 16 videos at 1.25x) - vectors, transformations, matrices, eigenvalues	https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab
1:45–3:15	3Blue1Brown "Essence of Calculus" (Chapters 1–7 at 1.25x) - derivatives, chain rule, integrals	https://www.youtube.com/playlist?list=PLZHQObOWTQDMsr9K-rj53DwVRMYO3t5Yr
3:15–4:15	3Blue1Brown "Neural Networks" series (Chapters 1–2) - what is a NN, gradient descent	https://www.3blue1brown.com/topics/neural-networks
4:15–5:00	Khan Academy exercises: matrix multiplication, derivatives, chain rule	https://www.khanacademy.org/math/linear-algebra

Day 4 (5 hrs) - Probability, statistics, and the math-ML bridge

Time	Activity	Resource
0:00–1:30	StatQuest: probability, distributions, Bayes' theorem, p-values (about 6 videos, search them in this channel)	https://www.youtube.com/playlist?list=PLblh5JKOoLUK0FLuzwntyYI10UQFUhsY9
1:30–2:30	StatQuest: linear regression, logistic regression, gradient descent	https://www.youtube.com/playlist?list=PLblh5JKOoLUIzaEkCLIUxQFjPIlapw8nU and https://www.youtube.com/watch?v=sDv4f4s2SB8 (for the playlist, only see the videos about the main topics, not coding in R or Anova stuff)
2:30–3:30	3Blue1Brown "Neural Networks" (Chapters 3–4) - backpropagation intuition + calculus	https://www.3blue1brown.com/topics/neural-networks
3:30–5:00	Skim Mathematics for Machine Learning book. Use NotebookLM (https://notebooklm.google.com) to understand faster - Chapter 2 (Linear Algebra notation) + Chapter 5 (Vector Calculus for optimization)	https://mml-book.github.io/book/mml-book.pdf

Reference to keep handy throughout the month:

- Mathematics for Machine Learning (free PDF): https://mml-book.github.io/book/mml-book.pdf
- Imperial College Coursera specialization (for deeper study later): https://www.coursera.org/specializations/mathematics-machine-learning

Days 5–7: ML and deep learning fundamentals (15 hours)

Dual-track: fast.ai for practical intuition, Google ML Crash Course for conceptual foundations.

Day 5 (5 hrs) - First ML models

Time	Activity	Resource
0:00–1:30	fast.ai Practical Deep Learning - Lesson 1 (train an image classifier, deploy it)	https://course.fast.ai/
1:30–3:00	fast.ai Lesson 2 (data cleaning, production deployment, Gradio)	https://course.fast.ai/
3:00–4:00	Kaggle Learn: Intro to Machine Learning (lessons 1–4, decision trees, validation)	https://www.kaggle.com/learn/intro-to-machine-learning
4:00–5:00	Google ML Crash Course: Introduction + Linear Regression modules	https://developers.google.com/machine-learning/crash-course

Day 6 (5 hrs) - Neural network foundations

Time	Activity	Resource
0:00–1:30	fast.ai Lesson 3 (neural net foundations, SGD, MNIST)	https://course.fast.ai/
1:30–3:00	Karpathy "Neural Networks: Zero to Hero" - Video 1: micrograd (backprop from scratch)	https://www.youtube.com/watch?v=VMj-3S1tku0&list=PLAqhIrjkxbuWI23v9cThsA9GvCAUhRvKZ&index=1 (if you're interested in the playlist: https://www.youtube.com/playlist?list=PLAqhIrjkxbuWI23v9cThsA9GvCAUhRvKZ)
3:00–4:30	Google ML Crash Course: Neural Networks + Training Neural Networks modules	https://developers.google.com/machine-learning/crash-course
4:30–5:00	Hands-on: Train a simple neural network in a Jupyter notebook using PyTorch	fast.ai notebooks

Day 7 (5 hrs) - DL concepts + first mini-project

Time	Activity	Resource
0:00–1:30	fast.ai Lesson 4 (NLP, tabular data, collaborative filtering)	https://course.fast.ai/
1:30–2:30	MIT 6.S191 Lecture 1: Introduction to Deep Learning (2025 edition)	https://introtodeeplearning.com/2025/index.html

Time	Activity	Resource
2:30–3:30	MIT 6.S191 Lecture 2: Deep Sequence Modeling (RNNs, attention)	https://introtodeeplearning.com/2025/index.html
3:30–5:00	**Mini-Project 1:** Train an image classifier on a custom dataset using fast.ai, deploy as a Gradio app on HF Spaces	https://huggingface.co/spaces

---

Week 2: Deep learning mastery and transformers (Days 8–14, 35 hours)

Days 8–10: Going deeper on neural networks (15 hours)

Day 8 (5 hrs)

Time	Activity	Resource
0:00–2:00	Karpathy Zero to Hero - Video 2: bigram language model (makemore pt 1)	https://www.youtube.com/watch?v=PaCmpygFfXo&list=PLAqhIrjkxbuWI23v9cThsA9GvCAUhRvKZ&index=2
2:00–4:00	Karpathy Zero to Hero - Video 3: MLP language model (makemore pt 2)	https://www.youtube.com/watch?v=TCH_1BHY58I&list=PLAqhIrjkxbuWI23v9cThsA9GvCAUhRvKZ&index=3
4:00–5:00	fast.ai Lesson 5 (from-scratch model, tabular data deep dive)	https://course.fast.ai/

Day 9 (5 hrs)

Time	Activity	Resource
0:00–2:00	Karpathy Zero to Hero - Video 4: activations, gradients, BatchNorm	https://www.youtube.com/watch?v=P6sfmUTpUmc&list=PLAqhIrjkxbuWI23v9cThsA9GvCAUhRvKZ&index=4
2:00–3:30	fast.ai Lessons 6–7 (random forests, collaborative filtering, convolutions)	https://course.fast.ai/

Time	Activity	Resource
3:30–5:00	MIT 6.S191 Lectures 3–4: CNNs + Generative AI foundations	https://introtodeeplearning.com/

Day 10 (5 hrs)

Time	Activity	Resource
0:00–2:00	Karpathy Zero to Hero - Video 5: manual backprop through an MLP	https://www.youtube.com/watch?v=q8SA3rM6ckI&list=PLAqhIrjkxbuWI23v9cThsA9GvCAUhRvKZ&index=5
2:00–3:30	fast.ai Lessons 8–9 (complete Part 1)	https://course.fast.ai/
3:30–5:00	Google ML Crash Course: Embeddings + LLM module	https://developers.google.com/machine-learning/crash-course/llm

Days 11–13: Transformers and foundation models (15 hours)

This is the core intellectual content of the entire bootcamp. **Every modern AI system is built on transformers.**

Day 11 (5 hrs) - Understanding attention and transformers

Time	Activity	Resource
0:00–0:45	Read Jay Alammar's "The Illustrated Transformer" (with diagrams)	https://jalammar.github.io/illustrated-transformer/
0:45–1:30	Read Jay Alammar's "The Illustrated GPT-2"	https://jalammar.github.io/illustrated-gpt2/
1:30–2:15	3Blue1Brown Chapter 5: "Transformers, the tech behind LLMs"	https://www.3blue1brown.com/lessons/gpt
2:15–3:00	3Blue1Brown Chapter 6: "Attention in transformers, step-by-step"	https://www.3blue1brown.com/lessons/attention
3:00–5:00	Karpathy Zero to Hero - Video 7: "Let's build GPT from scratch" (first half)	https://www.youtube.com/watch?v=kCc8FmEb1nY

Day 12 (5 hrs) - Building GPT from scratch

Time	Activity	Resource

Time	Activity	Resource
0:00–2:00	Karpathy "Let's build GPT" (second half - complete the implementation, code along in Colab)	https://colab.research.google.com/drive/1JMLa53HDuA-i7ZBmqV7ZnA3c_fvtXnx-
2:00–3:00	Karpathy Zero to Hero - Video 7: tokenizer (BPE from scratch)	https://www.youtube.com/watch?v=kCc8FmEb1nY&list=PLAqhIrjkxbuWI23v9cThsA9GvCAUhRvKZ&index=7
3:00–5:00	Hugging Face LLM Course - Chapter 1 (Transformer models, pipelines, what's inside the hub)	https://huggingface.co/learn/llm-course/chapter1/1

Day 13 (5 hrs) - Hugging Face ecosystem + nanoGPT

Time	Activity	Resource
0:00–2:00	Hugging Face LLM Course - Chapter 2 (Using Transformers library, models, tokenizers)	https://huggingface.co/learn/llm-course/chapter1/1
2:00–3:30	Hugging Face LLM Course - Chapter 3 (Fine-tuning a pretrained model)	Same course
3:30–5:00	Explore nanoGPT repo, read code, attempt a small training run on GPU	https://github.com/karpathy/build-nanogpt

Key reference materials:

- Sebastian Raschka, "Build a Large Language Model From Scratch" - code: https://github.com/rasbt/LLMs-from-scratch
- Harvard's "The Annotated Transformer": https://nlp.seas.harvard.edu/annotated-transformer/
- Lilian Weng's "The Transformer Family v2.0": https://lilianweng.github.io/posts/2023-01-27-the-transformer-family-v2/

Day 14: Mini-project + review (5 hours)

Time	Activity	Resource
0:00–3:00	**Mini-Project 2:** Fine-tune a small pre-trained model (e.g., DistilBERT) on a text classification task using HF Transformers + Trainer API	HF Course Chapter 3 notebooks
3:00–5:00	Deploy the fine-tuned model as a Gradio demo on Hugging Face Spaces	https://www.gradio.app/guides/quickstart

---

Week 3: LLM engineering - fine-tuning, RAG, and agents (Days 15–21, 35 hours)

Days 15–16: Fine-tuning LLMs (10 hours)

Day 15 (5 hrs) - LoRA, QLoRA, and modern fine-tuning

Time	Activity	Resource
0:00–1:00	Read Sebastian Raschka's "LoRA and DoRA from Scratch" blog post	https://magazine.sebastianraschka.com/p/lora-and-dora-from-scratch
1:00–1:30	Read Hugging Face blog: "Illustrating RLHF" (conceptual overview of alignment)	https://huggingface.co/blog/rlhf
1:30–3:30	**Hands-on:** Fine-tune Llama/Qwen with Unsloth using a provided Colab notebook (LoRA + QLoRA)	https://github.com/unslothai/unsloth + their Colab notebooks
3:30–5:00	Maxime Labonne's LLM Course - fine-tuning section (SFT concepts, dataset preparation)	https://github.com/mlabonne/llm-course

Day 16 (5 hrs) - Alignment: DPO, RLHF, and preference tuning

Time	Activity	Resource
0:00–1:30	Phil Schmid's "How to Fine-Tune LLMs in 2025" (QLoRA + TRL walkthrough)	https://www.philschmid.de/fine-tune-llms-in-2025
1:30–3:00	DPO hands-on: Phil Schmid's DPO alignment tutorial with TRL DPOTrainer	https://www.philschmid.de/dpo-align-llms-in-2024-with-trl
3:00–4:00	DeepLearning.AI RLHF short course (conceptual + hands-on)	https://www.deeplearning.ai/short-courses/reinforcement-learning-from-human-feedback/
4:00–5:00	Explore Hugging Face Alignment Handbook - read the Zephyr-7B recipe	https://github.com/huggingface/alignment-handbook

Key reference materials for fine-tuning:

- Hugging Face PEFT docs: https://huggingface.co/docs/peft/en/index
- Hugging Face TRL docs: https://huggingface.co/docs/trl/en/index
- Axolotl (for production fine-tuning): https://github.com/axolotl-ai-cloud/axolotl
- LoRA paper: https://arxiv.org/abs/2106.09685
- QLoRA paper: https://arxiv.org/abs/2305.14314
- DPO paper: https://arxiv.org/abs/2305.18290
- RLHF book (free PDF): https://rlhfbook.com/book.pdf

- Datasets for practice: Alpaca Cleaned (https://huggingface.co/datasets/yahma/alpaca-cleaned), Databricks Dolly (https://huggingface.co/datasets/databricks/databricks-dolly-15k), UltraFeedback (https://huggingface.co/datasets/HuggingFaceH4/ultrafeedback_binarized)

Days 17–18: RAG systems (10 hours)

Day 17 (5 hrs) - RAG fundamentals

Time	Activity	Resource
0:00–1:30	DeepLearning.AI "Building and Evaluating Advanced RAG" short course	https://www.deeplearning.ai/short-courses/building-evaluating-advanced-rag/
1:30–2:30	Read Weaviate's chunking strategies blog (comprehensive overview)	https://weaviate.io/blog/chunking-strategies-for-rag
2:30–4:00	**Hands-on:** Build a basic RAG pipeline with LangChain + ChromaDB + OpenAI embeddings (or Sentence Transformers all-MiniLM-L6-v2 for free option)	https://docs.langchain.com/oss/python/langchain/rag
4:00–5:00	Learn about embedding models: try free gemini or any opensource embedding model vs. Sentence Transformers	https://www.sbert.net/

Day 18 (5 hrs) - Advanced RAG techniques

Time	Activity	Resource
0:00–1:30	NirDiamant's RAG_Techniques repo - work through 3-4 notebooks (HyDE, reranking, fusion retrieval)	https://github.com/NirDiamant/RAG_Techniques
1:30–2:30	Read Anthropic's "Contextual Retrieval" blog post	https://www.anthropic.com/news/contextual-retrieval
2:30–3:30	Explore Microsoft GraphRAG - read docs and run getting started	https://github.com/microsoft/graphrag
3:30–5:00	**Hands-on:** Set up RAGAS evaluation for your RAG pipeline, run benchmarks	https://docs.ragas.io/en/stable/getstarted/rag_eval/

Key RAG references:

- LlamaIndex docs: https://developers.llamaindex.ai/python/framework/understanding/rag/
- ChromaDB: https://docs.trychroma.com/
- Pinecone: https://docs.pinecone.io/
- FAISS: https://github.com/facebookresearch/faiss
- Advanced RAG overview: https://www.meilisearch.com/blog/rag-types

Days 19–21: AI agents and agentic systems (15 hours)

This is the most critical section for agentic AI systems.

Day 19 (5 hrs) - Agent fundamentals

Time	Activity	Resource
0:00–1:30	Read Anthropic's "Building Effective Agents" blog post (**THE foundational guide**)	https://www.anthropic.com/research/building-effective-agents
1:30–2:30	Read Anthropic's "Writing effective tools for AI agents"	https://www.anthropic.com/engineering/writing-tools-for-agents
2:30–3:30	Hugging Face Agents Course - Unit 1 (agent fundamentals, ReAct pattern)	https://huggingface.co/learn/agents-course/en/unit0/introduction
3:30–5:00	**Hands-on:** Build your first agent with OpenAI Agents SDK (quickstart + tool use)	https://openai.github.io/openai-agents-python/

Day 20 (5 hrs) - Agent frameworks deep dive

Time	Activity	Resource
0:00–2:00	LangGraph tutorial - build a stateful, multi-step agent with tools	https://docs.langchain.com/oss/python/langgraph/overview
2:00–3:30	CrewAI quickstart - build a multi-agent crew with roles and goals	https://docs.crewai.com/en/quickstart
3:30–5:00	SmolAgents by Hugging Face - build a code agent that writes Python as actions	https://huggingface.co/docs/smolagents/en/index

Day 21 (5 hrs) - MCP, protocols, and mini-project

Time	Activity	Resource
0:00–1:30	Learn MCP (Model Context Protocol) - read spec overview, understand client-server architecture	https://modelcontextprotocol.io/
1:30–2:30	Read about A2A (Agent-to-Agent) protocol by Google	https://a2a-protocol.org/latest/
2:30–5:00	**Mini-Project 3:** Build a RAG-powered agent that can search documents AND use web tools. Use LangGraph or Google Agent SDK + ChromaDB for retrieval	Combine resources from Days 17–20

Key agent references:

- Anthropic "Effective context engineering": https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents
- OpenAI function calling docs: https://platform.openai.com/docs/guides/function-calling
- Andrew Ng's "Agentic AI" course on DeepLearning.AI: https://www.deeplearning.ai/courses/agentic-ai/
- DSPy (advanced - programmatic LLM pipelines): https://dspy.ai/
- Lilian Weng's "LLM-Powered Autonomous Agents": https://lilianweng.github.io/posts/2023-06-23-agent/

Week 4: Deployment, advanced topics, and capstone (Days 22–28, 35 hours)

Days 22–23: MLOps and deployment (10 hours)

Day 22 (5 hrs) - Containerization and API serving

Time	Activity	Resource
0:00–1:30	Docker for ML crash course (core concepts + ML Dockerfile)	https://machinelearningmastery.com/the-complete-guide-to-docker-for-machine-learning-engineers/
1:30–3:00	FastAPI for ML model serving (build a prediction endpoint + Swagger docs)	https://machinelearningmastery.com/step-by-step-guide-to-deploying-machine-learning-models-with-fastapi-and-docker/
3:00–4:00	Gradio deep dive - build an interactive demo for an LLM app	https://www.gradio.app/guides/quickstart
4:00–5:00	Deploy a Gradio demo to Hugging Face Spaces (free hosting)	https://huggingface.co/spaces

Day 23 (5 hrs) - LLM serving and experiment tracking

Time	Activity	Resource
0:00–1:30	vLLM: Set up high-throughput LLM serving on lab GPUs (OpenAI-compatible API)	https://docs.vllm.ai/en/latest/
1:30–2:30	Weights & Biases 101 - experiment tracking, logging, dashboards	https://www.wandb.courses/
2:30–3:30	MLflow quickstart - log experiments, register models	https://mlflow.org/docs/latest/index.html
3:30–5:00	**Hands-on:** Deploy a fine-tuned model via vLLM, track experiments in W&B, serve via FastAPI	Combine vLLM + W&B + FastAPI

Key deployment references:

- BentoML (easiest model packaging): https://docs.bentoml.com/
- Modal.com (serverless GPU): https://modal.com/
- Together AI (LLM inference API): https://docs.together.ai/
- Streamlit (ML dashboards): https://docs.streamlit.io/
- MLOps Zoomcamp (free comprehensive course): https://github.com/DataTalksClub/mlops-zoomcamp

Days 24–25: Advanced agentic systems (10 hours)

Day 24 (5 hrs) - Production agent patterns

Time	Activity	Resource
0:00–1:30	Anthropic "Demystifying evals for AI agents" - learn how to evaluate agent systems	https://www.anthropic.com/engineering/demystifying-evals-for-ai-agents
1:30–3:00	Build a multi-agent system with LangGraph (orchestrator + specialist agents)	https://docs.langchain.com/oss/python/langgraph/overview
3:00–4:00	Connect an agent to MCP servers (external tools, databases)	https://www.anthropic.com/engineering/code-execution-with-mcp
4:00–5:00	DeepLearning.AI "Building Code Agents with smolagents" (first half)	https://learn.deeplearning.ai/courses/building-code-agents-with-hugging-face-smolagents/

Day 25 (5 hrs) - Computer use, coding agents, and state of the art

Time	Activity	Resource
0:00–1:00	Survey the coding agent landscape: Claude Code, Codex CLI, SWE-Agent, OpenHands	https://github.com/princeton-nlp/SWE-agent
1:00–2:00	Explore Anthropic computer use API - understand screen-based agent capabilities	Anthropic API docs
2:00–3:30	DeepLearning.AI "Agentic AI" course by Andrew Ng (selected modules: reflection, planning)	https://www.deeplearning.ai/courses/agentic-ai/
3:30–5:00	**Hands-on:** Build an agentic RAG system that dynamically decides when to retrieve, search the web, or generate directly	Combine LangGraph + RAG pipeline

Days 26–28: Capstone project (15 hours)

The capstone is the most important part of the bootcamp. Interns work in teams of 3–5 to build a **complete AI system** that integrates skills from the entire month.

Recommended capstone project: AI Research Assistant

This project combines **every skill learned**: Python, ML concepts, transformers, fine-tuning, RAG, agents, and deployment.

Component	Skills Used
Document ingestion pipeline	Chunking strategies, embeddings, vector DB (ChromaDB/Qdrant)
RAG retrieval system	Semantic search, reranking, hybrid retrieval
Agent orchestration	LangGraph or OpenAI Agents SDK, tool use, ReAct pattern
LLM backbone	vLLM serving or API calls, prompt engineering
Web interface	Gradio or Streamlit frontend
Deployment	Docker containerization, FastAPI backend
Experiment tracking	W&B or MLflow logging

Day 26 (5 hrs) - Architecture and backend

Time	Activity	Resource
0:00–1:00	Team formation, project scoping, architecture whiteboarding	-
1:00–3:00	Build the RAG backend: ingestion pipeline, vector store, retrieval API	Previous RAG resources
3:00–5:00	Build the agent layer: tool definitions, orchestration logic, memory	Previous agent resources

Day 27 (5 hrs) - Integration and frontend

Time	Activity	Resource
0:00–2:00	Connect agent to RAG backend + web search tool + any MCP servers	MCP docs
2:00–3:30	Build Gradio/Streamlit UI with chat interface, document upload, source citations	https://www.gradio.app/
3:30–5:00	Dockerize the application, set up vLLM for LLM serving	Docker + vLLM docs

Day 28 (5 hrs) - Polish, evaluate, and present

Time	Activity	Resource
0:00–1:30	Evaluate system with RAGAS metrics + manual testing, fix failure modes	https://docs.ragas.io/

Time	Activity	Resource
1:30–3:00	Deploy to Hugging Face Spaces or lab server, write README documentation	-
3:00–4:00	Prepare 5-minute demo presentation	-
4:00–5:00	**Demo Day:** Each team presents their AI Research Assistant to the lab	-

---

The complete resource library organized by category

Foundational courses (must-complete)

Resource	URL	Hours
Kaggle Learn Python -> Pandas -> Viz -> Intro ML	https://www.kaggle.com/learn/python	16
3Blue1Brown Linear Algebra + Calculus + Neural Nets	https://www.3blue1brown.com/topics/neural-networks	10
fast.ai Practical Deep Learning Part 1	https://course.fast.ai/	14
Karpathy Neural Networks: Zero to Hero	https://karpathy.ai/zero-to-hero.html	18
Hugging Face LLM Course	https://huggingface.co/learn/llm-course/chapter1/1	20
Google ML Crash Course (2024 edition)	https://developers.google.com/machine-learning/crash-course	15
MIT 6.S191 Introduction to Deep Learning	https://introtodeeplearning.com/	20

Transformers and LLM architecture

Resource	URL	Format
Jay Alammar "The Illustrated Transformer"	https://jalammar.github.io/illustrated-transformer/	Blog
Jay Alammar "The Illustrated GPT-2"	https://jalammar.github.io/illustrated-gpt2/	Blog
3Blue1Brown Chapters 5–7 (Transformers + Attention)	https://www.3blue1brown.com/lessons/attention	Video
Karpathy "Let's build GPT from scratch"	https://www.youtube.com/watch?v=kCc8FmEb1nY	Video
nanoGPT / build-nanogpt repo	https://github.com/karpathy/build-nanogpt	Code

Resource	URL	Format
Raschka "Build an LLM from Scratch"	https://github.com/rasbt/LLMs-from-scratch	Book+Code
The Annotated Transformer (Harvard)	https://nlp.seas.harvard.edu/annotated-transformer/	Code

Fine-tuning LLMs

Resource	URL	Covers
Unsloth (2x faster, 70% less VRAM)	https://github.com/unslothai/unsloth	LoRA, QLoRA, SFT, DPO
Hugging Face PEFT library	https://huggingface.co/docs/peft/en/index	15+ PEFT methods
Hugging Face TRL (DPO/RLHF)	https://huggingface.co/docs/trl/en/index	SFT, DPO, PPO, GRPO
Axolotl (production fine-tuning)	https://github.com/axolotl-ai-cloud/axolotl	Full FT, LoRA, DPO
Phil Schmid's 2025 fine-tuning guide	https://www.philschmid.de/fine-tune-llms-in-2025	End-to-end tutorial
Alignment Handbook (Zephyr recipes)	https://github.com/huggingface/alignment-handbook	SFT->DPO pipeline
Maxime Labonne's LLM Course	https://github.com/mlabonne/llm-course	Complete LLM roadmap

RAG systems

Resource	URL	Best For
NirDiamant RAG_Techniques (20+ notebooks)	https://github.com/NirDiamant/RAG_Techniques	Hands-on practice
ChromaDB	https://docs.trychroma.com/	Learning + prototyping
LangChain RAG tutorial	https://docs.langchain.com/oss/python/langchain/rag	Framework approach
LlamaIndex RAG docs	https://developers.llamaindex.ai/python/framework/understanding/rag/	Data-centric RAG
Anthropic Contextual Retrieval	https://www.anthropic.com/news/contextual-retrieval	Advanced technique
Microsoft GraphRAG	https://github.com/microsoft/graphrag	Knowledge graph RAG

Resource	URL	Best For
RAGAS evaluation framework	https://docs.ragas.io/en/stable/	RAG evaluation
Sentence Transformers (free embeddings)	https://www.sbert.net/	Embedding models

AI agents and agentic AI

Resource	URL	Best For
Anthropic "Building Effective Agents"	https://www.anthropic.com/research/building-effective-agents	Must-read conceptual guide
OpenAI Agents SDK	https://openai.github.io/openai-agents-python/	Fastest to build
LangGraph	https://docs.langchain.com/oss/python/langgraph/overview	Complex stateful agents
CrewAI	https://docs.crewai.com/en/quickstart	Multi-agent teams
SmolAgents (Hugging Face)	https://huggingface.co/docs/smolagents/en/index	Code agents
MCP (Model Context Protocol)	https://modelcontextprotocol.io/	Agent-to-tool standard
A2A (Agent-to-Agent Protocol)	https://a2a-protocol.org/latest/	Agent-to-agent comms
HF Agents Course	https://huggingface.co/learn/agents-course/en/unit0/introduction	Structured learning
DeepLearning.AI Agentic AI (Andrew Ng)	https://www.deeplearning.ai/courses/agentic-ai/	Design patterns

Deployment and MLOps

Resource	URL	Best For

Resource	URL	Best For
vLLM (LLM serving)	https://docs.vllm.ai/en/latest/	High-throughput inference
FastAPI (API serving)	https://fastapi.tiangolo.com/	ML model endpoints
Gradio (ML demos)	https://www.gradio.app/	Interactive demos
Docker for ML guide	https://machinelearningmastery.com/the-complete-guide-to-docker-for-machine-learning-engineers/	Containerization
W&B (experiment tracking)	https://wandb.ai/	Tracking + dashboards
MLflow	https://mlflow.org/	Open-source tracking
Modal.com (serverless GPU)	https://modal.com/	Cloud deployment
MLOps Zoomcamp (free course)	https://github.com/DataTalksClub/mlops-zoomcamp	Full MLOps pipeline

---